CBC Digital Learning Platform

Stage D — Interaction & Behaviour Modelling

Unified Software Architectural Process

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Document Purpose

This document presents all UML interaction and behaviour models produced in Stage D of the Unified Software Architectural Process for the CBC Digital Learning Platform. It includes sequence diagrams, activity diagrams, and state diagrams, all directly traceable to requirements in the SRS v1.0.

Stage D: Interaction & Behaviour Modelling

Table of Contents

Table of Contents 1
1. Introduction 1
1.1 Purpose of Stage D 1
1.2 Scope and Coverage 1
2. Sequence Diagrams 1
2.1 SD-01 — Learner Login & Authentication 1
Scenario Description 1
Referenced Requirements 1
2.2 SD-02 — AI Tutor Query Flow 1
Scenario Description 1
Referenced Requirements 1
3. Activity Diagrams 1
3.1 AD-01 — Learner Content Access Activity 1
Process Description 1
Referenced Requirements 1
3.2 AD-02 — AI Tutor Interaction Activity 1
Process Description 1
Referenced Requirements 1
4. State Diagrams 1
4.1 STD-01 — Learner Session State Diagram 1
State Description 1
Referenced Requirements 1
4.2 STD-02 — Knowledge Post State Diagram 1
State Description 1
Referenced Requirements 1
4.3 STD-03 — AI Tutor Interaction State Diagram 1
State Description 1
Referenced Requirements 1
5. Stage D Requirements Traceability 1
6. Stage D Summary & Handoff to Stage E 1
6.1 Key Architectural Decisions Established in Stage D 1
6.2 Approval 1

1. Introduction

1.1 Purpose of Stage D

Stage D of the Unified Software Architectural Process — Interaction and Behaviour Modelling — bridges the structural domain model produced in Stage C with the architectural design that will follow in Stage E. Where Stage C defined what objects exist in the system and their relationships, Stage D defines how those objects interact over time, how processes flow through the system, and how individual objects transition between states in response to events.

This document presents all UML behavioural models for the CBC Digital Learning Platform, covering the three diagram types specified for this stage:

- Sequence Diagrams — showing time-ordered interactions between system objects for specific use case flows
- Activity Diagrams — showing the procedural flow of key system processes including decision points and parallel actions
- State Diagrams — showing the lifecycle of key domain objects: the learner session, the knowledge post, and the AI tutor interaction

1.2 Scope and Coverage

The interactions modelled in this document were selected based on their architectural significance to the platform. Every diagram traces directly to one or more functional or non-functional requirements from the SRS v1.0. The following table summarises the deliverables in this stage:

Diagram ID	Title	Primary SRS References
SD-01	Learner Login & Authentication	FR-L-01, FR-L-02, NFR-SEC-02, NFR-SEC-03, NFR-SEC-04
SD-02	AI Tutor Query Flow	FR-L-05, FR-AI-01 to FR-AI-05, NFR-P-02, NFR-SEC-05
AD-01	Learner Content Access Activity	FR-L-03, FR-L-04, NFR-P-01, NFR-P-03, NFR-O-01 to NFR-O-03
AD-02	AI Tutor Interaction Activity	FR-L-05, FR-AI-01 to FR-AI-06, NFR-O-03, NFR-P-02
STD-01	Learner Session State Diagram	FR-L-02, NFR-SEC-03, NFR-SEC-05, NFR-AC-03, NFR-O-01
STD-02	Knowledge Post State Diagram	FR-L-06, FR-L-07, FR-L-10, FR-L-11, NFR-SEC-05
STD-03	AI Tutor Interaction State Diagram	FR-L-05, FR-AI-01 to FR-AI-05, NFR-A-03

2. Sequence Diagrams

Sequence diagrams model the time-ordered exchange of messages between system participants (actors and components) for specific scenarios. Each diagram shows the lifeline of every participant, the messages exchanged, activation periods, and any conditional alternative flows (alt fragments).

2.1 SD-01 — Learner Login & Authentication

Scenario Description

This sequence models the complete login flow from the moment a learner submits credentials to the delivery of a role-appropriate dashboard. The interaction spans five system participants: the Learner (actor), the Web Client (browser), the API Gateway (REST backend), the Auth Service (handles bcrypt comparison and JWT generation), and the User Store (database).

A critical alt fragment governs the conditional logic: if the submitted password matches the stored bcrypt hash, a signed JWT is generated and returned to the client; if not, a 401 Unauthorized response is returned. JWT is stored in session storage — not persisted across browser sessions — supporting the shared device requirement (NFR-AC-03). The final dashboard rendered is role-determined (learner / teacher / MoES), enforcing NFR-SEC-04.

Referenced Requirements

Req. ID	Requirement
FR-L-01	System shall allow a learner to register an account using username and password.
FR-L-02	System shall allow a learner to log in and log out securely.
NFR-SEC-02	All learner data shall be encrypted at rest in the database.
NFR-SEC-03	Authentication shall use secure password hashing (bcrypt). Plain-text passwords never stored.
NFR-SEC-04	Access control shall be role-based: learner, teacher, MoES roles strictly separated.

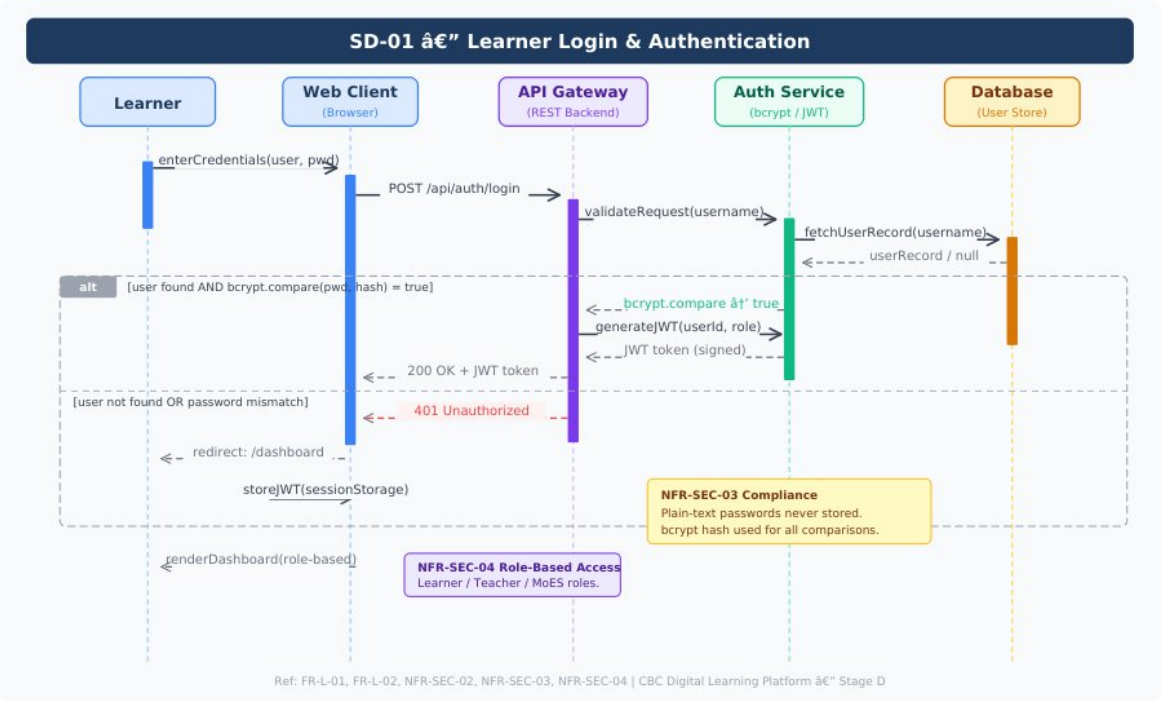

Figure SD-01: Learner Login & Authentication Sequence Diagram

SD-01 | FR-L-01, FR-L-02, NFR-SEC-03, NFR-SEC-04

2.2 SD-02 — AI Tutor Query Flow

Scenario Description

This sequence models the full lifecycle of a learner submitting a question to the AI Tutor — from the initial UI interaction to the delivery of a contextualised response. Six participants are involved: the Learner, the Web Client (AI Tutor inline panel), the API Gateway (handles JWT validation and rate limiting), the MCP Server (queries the NCDC knowledge base), the AI Model (Claude via API, generating the grounded response), and the Session Log (database recording interaction history).

The central design decision modelled here is the MCP-first retrieval strategy: the system queries the domestic NCDC knowledge base before involving the AI model. This grounds all responses in Uganda-specific curriculum content (FRAI-02, FRAI-03). The alt fragment captures the two outcomes: if the MCP server returns relevant context chunks, the AI model generates a grounded response; if no relevant content is found, the system invokes FRAI-05 and returns a graceful "outside knowledge base" message rather than generating a potentially inaccurate response.

All AI interactions are logged to the Session Log for future analytics, but remain private by default (NFR-SEC-05). NFR-P-02 governs the 5-second response time target.

Referenced Requirements

Req. ID	Requirement
FR-L-05	System shall allow a learner to submit an academic question to the AI tutor and receive a contextualised response.
FRAI-01	System shall embed an AI tutor accessible inline within the learner content experience.
FRAI-02	AI tutor shall answer academic questions grounded in Uganda's CBC curriculum content.
FRAI-03	AI tutor shall query domestic Ugandan data through MCP servers to contextualise responses.
FRAI-05	AI tutor shall indicate when a question falls outside its knowledge base — no inaccurate responses.
NFR-P-02	AI tutor shall return a response within 5 seconds under normal load.
NFR-SEC-05	All learner activity and AI interaction data shall be private by default.

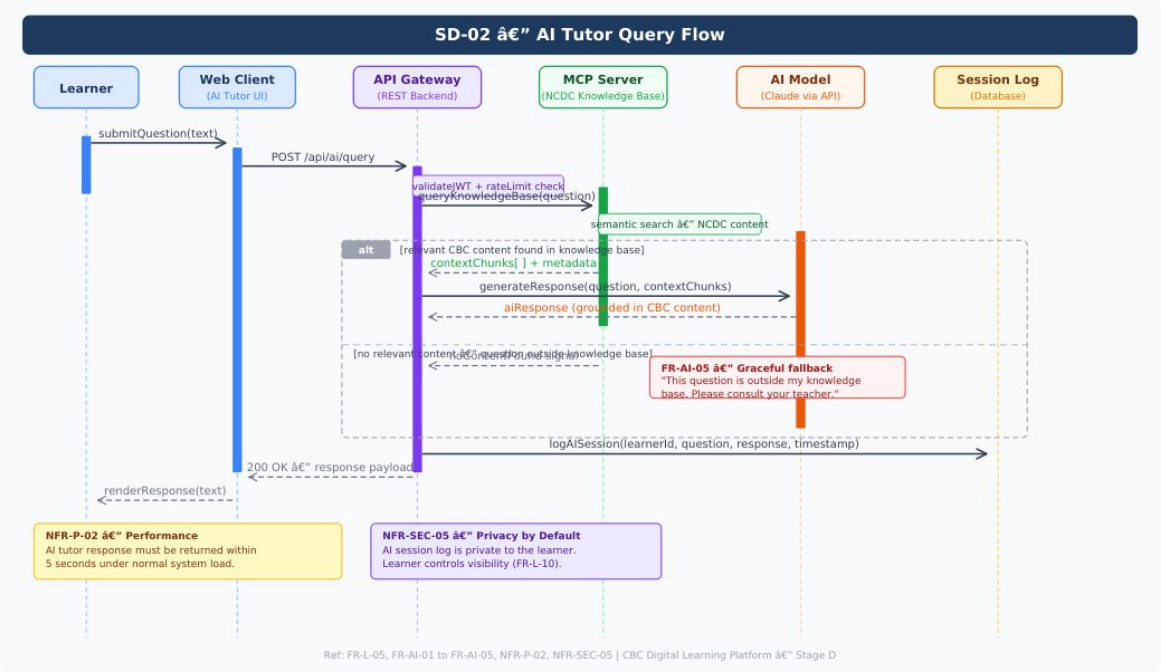

Figure SD-02: AI Tutor Query Flow Sequence Diagram

SD-02 | FR-L-05, FRAI-01 to FRAI-05, NFR-P-02, NFR-SEC-05

3. Activity Diagrams

Activity diagrams model procedural process flows through the system, showing actions, decision nodes (diamonds), forks and joins, and swimlane partitions that clarify which system participant is responsible for each action. They are particularly effective for capturing conditional and parallel behaviour that would be difficult to express in a sequence diagram.

3.1 AD-01 — Learner Content Access Activity

Process Description

This activity diagram models the process by which a learner accesses curriculum content on the platform. Three swimlanes partition responsibility: the Learner/Browser, the Backend/Cache, and the Database/Storage. The critical decision node is the connectivity check at the beginning of the flow: if the learner is online, the system proceeds through the full request-query-return-render path, with the backend caching lesson content in the browser for future offline use. If the learner is offline, the system serves previously cached content directly from the browser.

The diagram captures the NFR-P-01 constraint (content loads within 3 seconds on 3G), NFR-P-03 (media compression and lazy loading), and the full NFR-O-01 offline flow. It also shows the optional "mark for offline" action, where the learner explicitly downloads a lesson.

Referenced Requirements

Req. ID	Requirement
FR-L-03	System shall present CBC curriculum content organised by subject, class level, and competency.
FR-L-04	System shall allow a learner to access previously loaded lessons while offline.
NFR-P-01	System shall load core curriculum content within 3 seconds on a 3G mobile connection.
NFR-P-03	System shall optimise all media assets for low-bandwidth delivery through compression and lazy loading.
NFR-O-01	A learner shall be able to read previously loaded lessons without an active internet connection.
NFR-O-02	System shall clearly indicate when the learner is in offline mode and which features are unavailable.

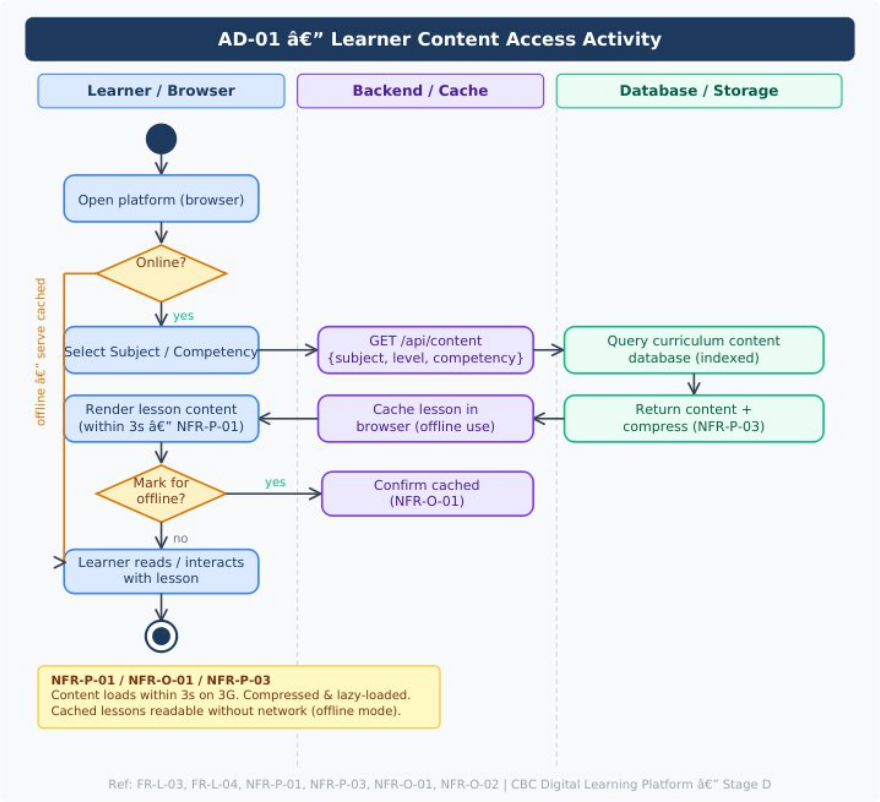

Figure AD-01: Learner Content Access Activity Diagram

AD-01 | FR-L-03, FR-L-04, NFR-P-01, NFR-P-03, NFR-O-01, NFR-O-02

3.2 AD-02 — AI Tutor Interaction Activity

Process Description

This activity diagram models the full AI Tutor interaction process across three swimlanes: the Learner, the Backend/MCP, and the AI Model & Session Log. It begins with an online connectivity check — a hard requirement since the AI tutor cannot function offline (NFR-O-03). If offline, the learner receives a clear message indicating this feature is unavailable.

When online, the flow proceeds through JWT validation, MCP knowledge base retrieval (semantic search against NCDC content), and a critical decision fork on whether relevant content was found. The yes-branch triggers AI response generation; the no-branch triggers the FR-AI-05 graceful fallback message — "This question is outside my knowledge base. Please consult your teacher." All interactions conclude with session logging, and the response is streamed to the learner UI.

Referenced Requirements

Req. ID	Requirement
FRAI-03	AI tutor shall query domestic Ugandan data through MCP servers to contextualise responses with local knowledge.
FRAI-04	AI tutor shall explain concepts, summarise topics, and generate relevant examples.
FRAI-05	AI tutor shall indicate when a question falls outside its knowledge base rather than generating inaccurate responses.
NFR-O-03	AI tutor queries shall require an active internet connection and shall be clearly marked as online-only.
NFR-P-02	AI tutor shall return a response within 5 seconds under normal load.

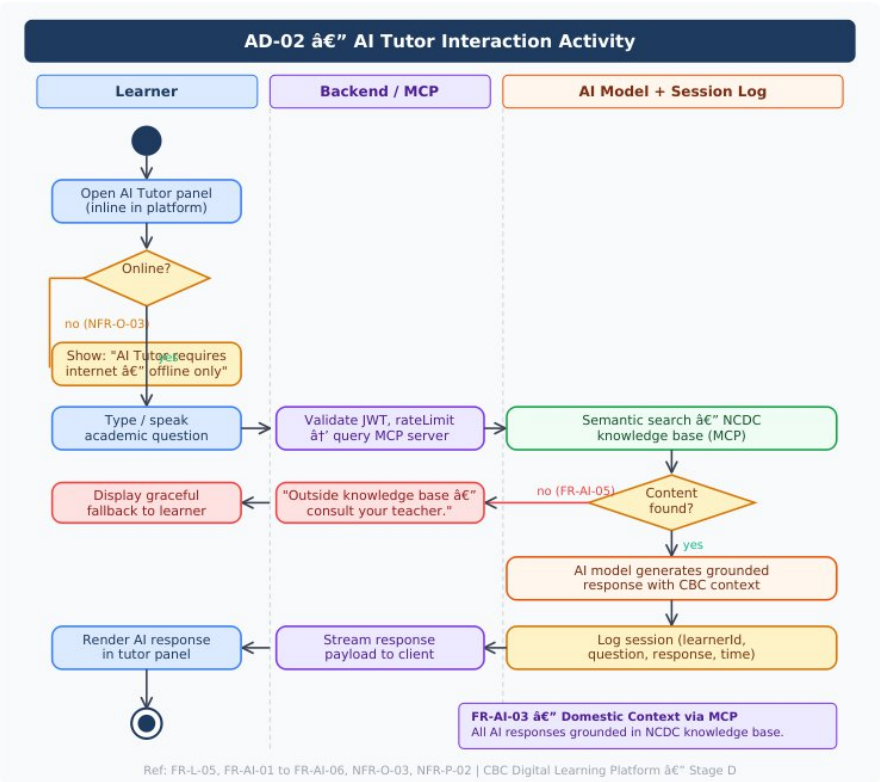

Figure AD-02: AI Tutor Interaction Activity Diagram

AD-02 | FR-L-05, FRAI-01 to FRAI-06, NFR-O-03, NFR-P-02

4. State Diagrams

State diagrams (UML statechart diagrams) model the lifecycle of a single object by defining its possible states, the events that trigger transitions between states, and any entry/exit/do behaviours associated with each state. Three key domain objects were selected for modelling based on their architectural complexity and system significance.

4.1 STD-01 — Learner Session State Diagram

State Description

This diagram models the lifecycle of a learner's authenticated session. The session object begins in the Unauthenticated state and transitions to Authenticating when the learner submits credentials. A valid JWT transitions it to Active; invalid credentials return it to Unauthenticated. From Active, the session can transition to Offline (on network loss — content remains accessible via cache), Idle (on 20 minutes of inactivity), or directly to the end state on logout.

The Idle state incorporates a countdown timer. If the learner resumes activity within the idle window, the session returns to Active. If the timeout expires (30 minutes), the session transitions to Expired, which clears the JWT, notifies the learner, and redirects to the login page. This design satisfies NFR-AC-03's shared device requirement by ensuring sessions do not persist unintentionally on shared school computers.

Referenced Requirements

Req. ID	Requirement
FR-L-02	System shall allow a learner to log in and log out securely.
NFR-SEC-03	Authentication shall use secure password hashing. Plain-text passwords never stored.
NFR-SEC-05	All learner activity and AI interaction data shall be private by default.
NFR-AC-03	Platform shall support quick session switching to accommodate shared device usage in schools.
NFR-O-01	Learner shall be able to read previously loaded lessons without an active internet connection.

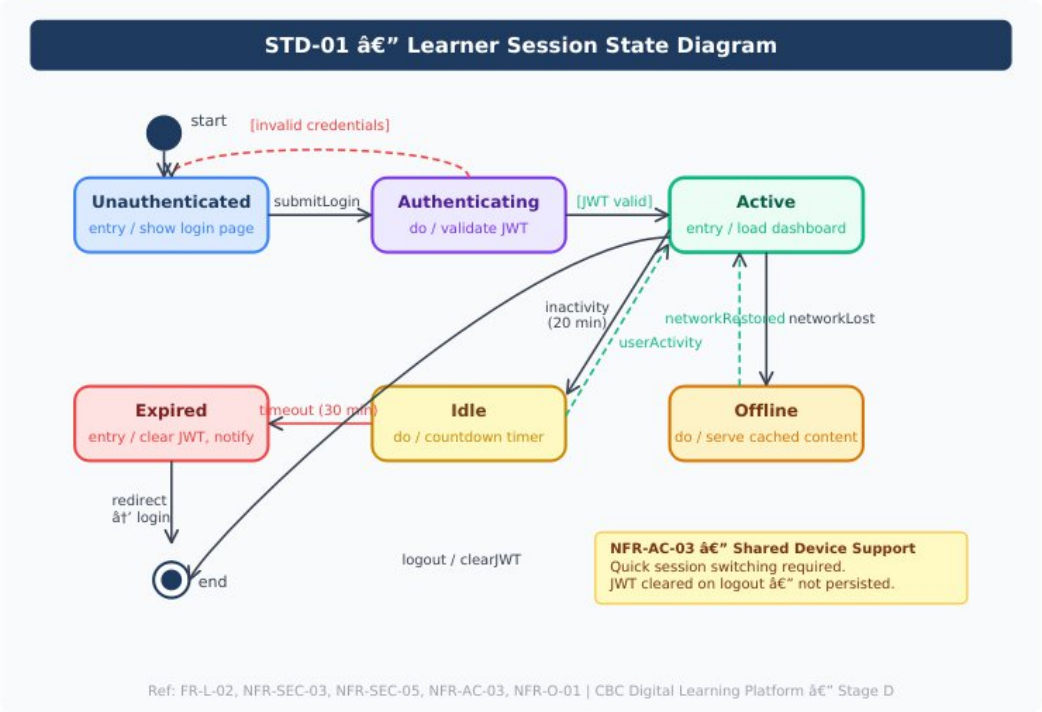

Figure STD-01: Learner Session State Diagram

STD-01 | FR-L-02, NFR-SEC-03, NFR-SEC-05, NFR-AC-03, NFR-O-01

4.2 STD-02 — Knowledge Post State Diagram

State Description

This diagram models the lifecycle of a learner's knowledge post — a piece of content (insight, idea, or research contribution) shared on the platform. A post begins as a Draft (auto-saved every 60 seconds) and transitions to Pending on submission, where an automated content policy check occurs. Posts that pass transition to Published (private by default, visible only to the creating learner — enforcing NFR-SEC-05 and FR-L-10). The learner can then set visibility to Shared, making the post visible to peers, teachers, or the public.

From Shared, a post may be Flagged by other users for content review. A moderation review can either clear the flag (returning to Shared) or confirm a violation (transitioning to Archived). Both the learner and moderators can also directly archive a post, which hides it from feeds while retaining the data. This lifecycle design enforces the platform's privacy-by-default principle at every stage.

Referenced Requirements

Req. ID	Requirement
FR-L-06	System shall allow a learner to post insights, ideas, and research contributions to the platform.
FR-L-07	System shall allow a learner to share knowledge and interact with posts from other learners nationally.
FR-L-10	System shall allow a learner to set the visibility of their posts (private by default).
FR-L-11	System shall allow a learner to choose to make specific posts visible to peers, teachers, or the public.
NFR-SEC-05	All learner activity and posts shall be private by default. Learners control visibility.

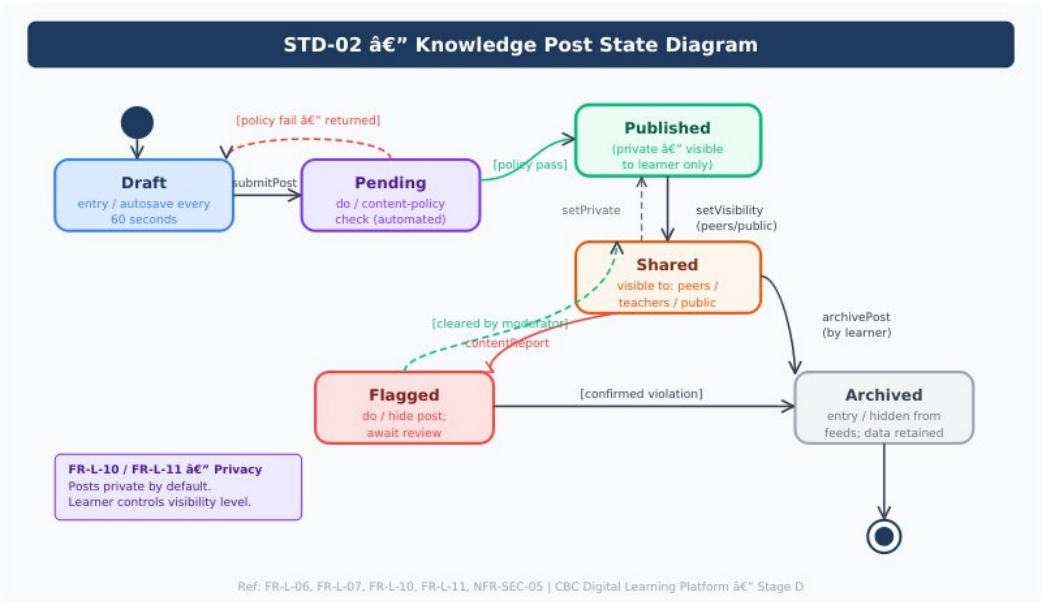

Figure STD-02: Knowledge Post State Diagram

STD-02 | FR-L-06, FR-L-07, FR-L-10, FR-L-11, NFR-SEC-05

4.3 STD-03 — AI Tutor Interaction State Diagram

State Description

This diagram models the internal lifecycle of a single AI Tutor interaction session object. The interaction begins in the Idle state (prompt input visible). On question submission, it transitions to Querying MCP, where a semantic search is performed against the NCDC knowledge base. The outcome of this search determines the next transition: if content is found, the object moves to Generating, where the AI model builds its response; if no content is found, it transitions directly to Out of Scope, triggering the FR-AI-05 graceful fallback.

From the Generating state, the interaction may transition to Responding on a successful AI response, or to Error if the API call fails (triggering a retry option). From Responding, the learner may submit a follow-up question (returning to Idle for a new cycle) or close the session (terminal state). The Error state also supports a retry transition back to Idle. NFR-A-03 governs the degradation scenario: if the entire AI layer is unavailable, the platform must continue serving core content without the tutor.

Referenced Requirements

Req. ID	Requirement
FR-L-05	System shall allow a learner to submit an academic question to the AI tutor and receive a contextualised response.
FRAI-04	AI tutor shall explain concepts, summarise topics, and generate relevant examples for the learner.
FRAI-05	AI tutor shall indicate when a question falls outside its knowledge base — shall not generate inaccurate responses.
NFR-A-03	System shall degrade gracefully — if the AI layer is unavailable, core content access shall remain functional.

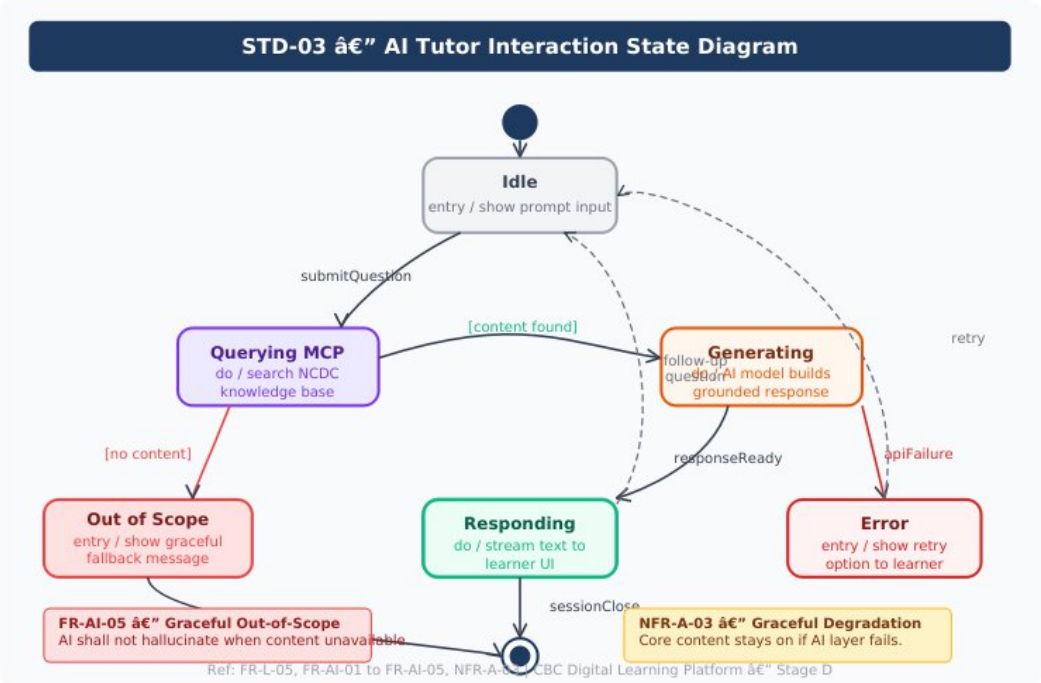

Figure STD-03: AI Tutor Interaction State Diagram

STD-03 | FR-L-05, FRAI-01 to FRAI-05, NFR-A-03

5. Stage D Requirements Traceability

The following matrix confirms that every diagram produced in Stage D traces to at least one validated requirement from the SRS v1.0, and that the key behavioural requirements from Sections 4 and 5 of the SRS are represented in at least one Stage D model.

Requirement	Description (abbreviated)	Covered In
FR-L-01	Learner registration	SD-01
FR-L-02	Learner login / logout	SD-01, STD-01
FR-L-03	CBC content by subject/level	AD-01
FR-L-04	Offline lesson access	AD-01, STD-01
FR-L-05	AI tutor question/response	SD-02, AD-02, STD-03
FR-L-06	Learner post contributions	STD-02
FR-L-07	National knowledge sharing	STD-02
FR-L-10	Post visibility (private default)	STD-02
FR-L-11	Learner sets post visibility	STD-02
FRAI-01	AI tutor inline in platform	SD-02, AD-02
FRAI-02	AI grounded in CBC content	SD-02, AD-02
FRAI-03	AI queries MCP / NCDC KB	SD-02, AD-02, STD-03
FRAI-04	AI explains concepts & examples	STD-03
FRAI-05	Graceful out-of-scope response	SD-02, AD-02, STD-03
NFR-P-01	Content loads < 3s on 3G	AD-01
NFR-P-02	AI responds < 5s	SD-02, AD-02
NFR-P-03	Media compression / lazy load	AD-01
NFR-A-03	Graceful degradation if AI fails	STD-03
NFR-SEC-03	bcrypt hashing — no plain-text	SD-01
NFR-SEC-04	Role-based access control	SD-01
NFR-SEC-05	Learner data private by default	SD-02, STD-02
NFR-O-01	Offline lesson reading	AD-01, STD-01
NFR-O-02	Offline mode indication	AD-01
NFR-O-03	AI requires internet connection	AD-02
NFR-AC-03	Shared device session switching	STD-01

6. Stage D Summary & Handoff to Stage E

Stage D is now complete. All seven behavioural models have been produced, reviewed against the SRS v1.0 requirements, and the full requirements traceability matrix confirms coverage of all key functional and non-functional requirements assigned to this stage.

6.1 Key Architectural Decisions Established in Stage D

The interaction models have surfaced the following decisions that must be carried forward into Stage E (Architectural Design):

- MCP-First Retrieval Architecture: the AI Tutor always queries the NCDC knowledge base before invoking the AI model API. Stage E must define the MCP server component, its interface to the backend, and the knowledge base schema.
- Stateless Session with JWT: authentication is stateless — the backend holds no session state; the JWT token is the session. Stage E must specify the JWT signing infrastructure, expiry policy, and token refresh strategy.
- Offline-First Content Cache: the browser acts as a secondary content store for offline scenarios. Stage E must define the caching technology (e.g. Service Workers / IndexedDB), the sync strategy when connectivity resumes, and the max offline cache size.
- Role-Based Access on Every Request: JWT payloads carry role claims; the API Gateway validates these on every request. Stage E must design the RBAC middleware layer.
- Graceful AI Degradation: the platform must remain fully functional if the AI layer is unavailable. Stage E must confirm the AI layer is a separate, independently deployable component with circuit-breaker logic.

6.2 Approval

This Stage D document represents the agreed baseline for interaction and behaviour modelling. Modifications to any diagram after approval must follow the formal change request process established in the SRS v1.0 and be assessed for impact on Stage E architectural decisions.

Role	Name	Signature / Date
Project Supervisor	_____	_____
Software Architect / Group Leader	_____	_____
Reviewer	_____	_____

— End of Document —